Lesson 28: Probability & Venn Diagrams — Practice Worksheet

Name: ______________________________ Date: ____________

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. A fair die: P(rolling a 4)?

2. A fair coin: P(heads)?

3. A die: P(an even number)?

4. If $P(A) = 0.3$, find $P(A')$.

5. A die: P(not a 6)?

6. A bag of 3 red, 7 blue: P(red)?

Section 2 — Core practice

7. $P(A) = 0.6$, $P(B) = 0.3$, $P(A \cap B) = 0.1$. Find $P(A \cup B)$.

8. A die: P(a number less than 3)?

9. Mutually exclusive A, B: $P(A) = 0.4$, $P(B) = 0.5$. Find $P(A \cup B)$.

10. If P(rain) = 0.35, find P(no rain).

11. $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cup B) = 0.7$. Find $P(A \cap B)$.

12. A bag of 4 red, 6 green: P(green)?

Section 3 — Challenge & exam-style

13. In a class of 30, 16 study French, 11 Spanish, 4 both. How many study neither?

14. Using the data above, find P(studies French only).

15. $P(A)=0.5$, $P(B)=0.6$ and A, B independent. Find $P(A\cap B)$.

16. Two events have $P(A)=0.7$, $P(B)=0.4$, $P(A\cup B)=0.9$. Are they mutually exclusive? Explain.